

Technical Slide Deck

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On

EMCAD: Efficient Multi-scale Convolutional Attention Decoding for
Medical Image Segmentation

Research Problem & EMCAD's Central Idea

Why EMCAD?: Transformer-based encoders provide strong global representations for medical segmentation, but the decoder can become the computational bottleneck.

EMCAD's idea: Replace expensive decoder operations with an efficient multi-scale convolutional attention decoder that:

- progressively refines hierarchical encoder features;
- captures multi-scale spatial context;
- selectively fuses skip-connection features;
- performs upsampling with depth-wise convolution;
- provides multi-stage supervision.
- The key design choice Use depth-wise convolutions at multiple scales instead of relying on expensive standard convolutions.

This gives EMCAD its central accuracy–efficiency trade-off.

Research question:

Can a lightweight convolutional decoder preserve or improve segmentation accuracy while substantially reducing decoder complexity?

EMCAD's Contribution

Table 1: 3 main contributions of EMCAD

MSCAM — Multi-Scale Convolutional Attention Module	LGAG — Large-Kernel Grouped Attention Gate	EUCB — Efficient Up-Convolution Block
CAB → SAB → MSCB • Channel attention emphasizes relevant feature channels. • Spatial attention captures local contextual information. • MSCB refines features using multi-scale depth-wise convolutions. • The multi-scale kernels are used in parallel.	Instead of the conventional attention gate: 3×3 grouped convolutions → attention coefficients → feature gating This allows larger local context while keeping computation low.	Upsample ×2 → 3×3 depth-wise convolution → BN/ReLU → 1×1 convolution The important point is that the expensive 3×3 operation is replaced by a depth-wise convolution.

Overall decoder

PVTv2 hierarchical features



MSCAM refinement



EUCB upsampling + LGAG skip fusion



Four segmentation heads / multi-stage supervision

Evidence That the Design Choices Matter

Table 2: Details of Multi-Scale Kernels

Depth-wise kernels	ClinicDB Dice
[1]	94.81
[1,3]	95.13
[1,3,5]	95.21
[3,5,7]	95.03
[1,3,5,7,9]	95.07

PVT-EMCAD-B2 on ClinicDB:

- Sequential: **95.15 $\pm$ 0.3**
- Parallel: **95.21 $\pm$ 0.2**

Table 3: LGAG vs conventional Attention Gate

	Params	FLOPs	Dice
AG	124.68K	61.68M	83.51
LGAG	11.01K	10.47M	83.63

What the Original Paper Established

The paper evaluates **EMCAD** on **12 public datasets** spanning six medical segmentation tasks.

For the ten binary segmentation datasets:

PVT-EMCAD-B2: 91.10% average Dice, 26.76M parameters, 5.6G FLOPs.

ClinicDB — relevant to my replication, **PVT-EMCAD-B2: 95.21% Dice**

The paper reports that EMCAD outperforms the corresponding **CASCADE decoder** while substantially reducing decoder complexity.

For PVTv2-B2:

- 80.3% fewer FLOPs
- 79.4% fewer parameters
- 0.85 percentage-point Dice improvement over CASCADE

EMCAD attempts to move the Pareto frontier: maintaining/improving segmentation quality while substantially reducing decoder computational cost.

My Replication: ClinicDB

Table 4: Comparison of Original and Replication Details

	Original	Replication
Model	PVT-EMCAD-B2	PVT-EMCAD-B2
Dataset	ClinicDB	ClinicDB
Input	352×352	352×352
Epochs	200	200
Batch size	16	4
GPU	RTX A6000 48 GB	Tesla T4
Runs	5	1

Replication Findings

Table 5: Replication Model's Performance Metrics

Metric	My result
Dice	95.02%
IoU	90.72%
Sensitivity	95.48%
Specificity	99.69%
Precision	94.87%
HD95	8.108
Test images	62